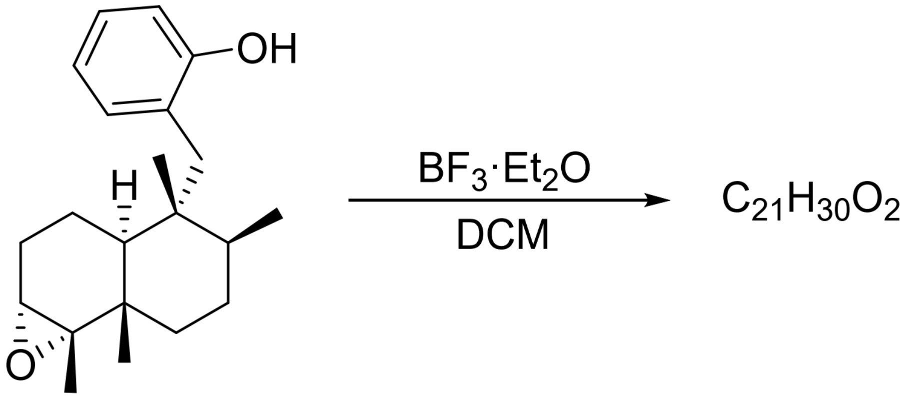
Question

The following reaction undergoes two carbocation rearrangement processes to finally obtain product **A**. Regarding the final product **A** of the reaction, which of the following statements are correct:



[H][C@]12[C@]([C@H](CC[C@]1(C)[C@]1([C@@H](CC2)O1)C)C)(C)CC1C(=CC=CC=1)O>BF_3·Et_2O>[A], where the molecular formula of A is $C_{21}H_{30}O_2$, and the reaction solvent is DCM

A. **A** has two rings

B. **A** The stereochemistry of the carbon atom attached to the hydroxyl group in is S.

C. **A** has 5 stereocenters, of which 2 have R configuration and 3 have S configuration.

D. **A** has 5 stereocenters, of which 3 have R configuration and 2 have S configuration.

E. **A** has 6 stereocenters, of which 3 are R configuration and 3 are S configuration.

F. **A** has 6 stereocenters, of which 2 have R configuration and 4 have S configuration.

G. All of the above options are incorrect.

Answer

Correct Answer: C

Detailed Explanation

The epoxide group of the substrate undergoes ring-opening under the catalysis of $BF_3 \cdot Et_2O$ (possibly via an oxonium ion mechanism), generating a tertiary carbocation.

CHECKPOINT 1 PTS

The epoxide group of the substrate undergoes ring-opening under the catalysis of $BF_3 \cdot Et_2O$, generating a tertiary carbocation

According to the question, two carbocation rearrangements are involved. First, the methyl group on the adjacent quaternary carbon undergoes a concerted sigmatropic migration to generate a tertiary carbocation at the bridgehead.

CHECKPOINT 1 PTS

The methyl group on the quaternary carbon undergoes a concerted sigmatropic migration to generate a tertiary carbocation at the bridgehead

Subsequently, a 1,2-sigmatropic hydride shift occurs. Finally, due to steric hindrance, the phenolic hydroxyl group attacks the carbocation in a concerted manner with proton transfer, ultimately resulting in a cis-fused ring structure.

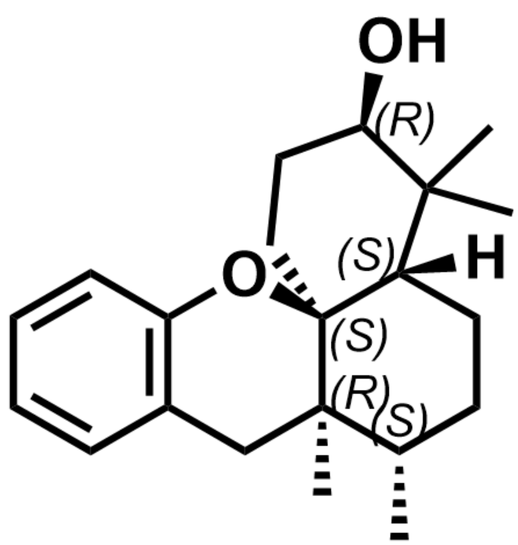
CHECKPOINT 1 PTS

Subsequently, a 1,2-sigmatropic hydride shift occurs

CHECKPOINT 1 PTS

Due to steric hindrance, the phenolic hydroxyl group attacks the carbocation in a concerted manner with proton transfer, ultimately resulting in a cis-fused ring structure

The structure of the final product **A** is as follows:



[H][C@@]12CC[C@H](C)[C@]3(C)[C@]1(OC4=C(C3)C=CC=C4)CC[C@@H](O)C2(C)C，which has 5 chiral centers, with 2 R configurations and 3 S configurations

Therefore, **A** has three rings.

CHECKPOINT 1 PTS

A has three rings, option A is incorrect

The stereoconfiguration of the carbon attached to the hydroxyl group is R.

CHECKPOINT 1 PTS

The stereoconfiguration of the carbon attached to the hydroxyl group is R, option B is incorrect

A has 5 stereocenters, with 2 R configurations and 3 S configurations.

CHECKPOINT 2 PTS

A has 5 stereocenters, with 2 R configurations and 3 S configurations, option C is correct, options D, E, and F are incorrect

In summary, the correct answer is C.

CHECKPOINT 1 PTS

The correct answer is C